DATA CLEANING — CHANGE LOG

Project 1 | DecodeLabs
Industrial Training Kit
Batch 2026

Prepared by: **Md Zulkar** Dataset: **E-commerce Orders CSV** Date: **16 May 2026** Tool: **Python / Pandas**

Change Log — Detailed Audit Trail

Change ID	Column(s)	Description	Impact / Outcome	Rationale	Status
CR001	CouponCode	Found 309 nulls in CouponCode. Instead of dropping these rows, filled them with 'No Coupon' since most customers simply didn't use one.	309 rows kept. Didn't lose any order records.	Dropping these rows would've removed valid orders just because no coupon was used — that didn't make sense.	Resolved
CR002	All columns	Ran a duplicate check across the full dataset using drop_duplicates(). Checked both full-row duplicates and OrderID-level duplicates separately.	No duplicates found either way. Dataset looks clean on this front.	Duplicate orders would mess up revenue totals and order counts badly.	Verified
CR003	Product, PaymentMethod, OrderStatus, ReferralSource, ShippingAddress	Noticed some inconsistency risk in text columns — applied strip() to remove extra spaces and title() for uniform casing on all categorical fields.	Casing is now consistent. Avoids grouping errors later during analysis.	Something like 'credit card' and 'Credit Card' would count as two separate payment methods otherwise.	Resolved

Change ID	Column(s)	Description	Impact / Outcome	Rationale	Status
CR004	CouponCode	Converted all non-null coupon codes to uppercase using str.upper(). Codes like 'save20' and 'SAVE20' were referring to the same coupon.	All active coupon entries are now uppercase and consistent.	Coupon matching in reports would break if the same code appeared in different cases.	Resolved
CR005	Date	Date column was stored as plain text (dtype: object). Values were already in YYYY-MM-DD format so just converted using pd.to_datetime() without any reformatting.	Converted cleanly — zero NaT values after conversion. Date arithmetic now works.	Can't do any time-based filtering or monthly grouping while it's sitting as a string.	Resolved
CR006	Quantity, UnitPrice, TotalPrice	Checked all three numeric columns for symbols, text or formatting issues. Quantity came as int64, UnitPrice and TotalPrice as float64 — all fine as-is.	Nothing to fix here. Skipped the conversion step.	Worth checking anyway — raw exports often sneak in things like '120000 INR' or '■500' whichbreak calculations.	Skipped
CR007	Quantity, UnitPrice, TotalPrice	Looked for values that don't make business sense — negative quantities, zero or negative prices, and orders dated in the future.	Zero issues found across all three checks.	A single negative quantity slipping through would quietly throw off total sales figures.	Verified

Change ID	Column(s)	Description	Impact / Outcome	Rationale	Status
CR008	All 14 columns	All column names were in CamelCase (e.g. OrderID, TotalPrice). Wrote a small function to convert them to snake_case — order_id, total_price, etc.	All 14 columns renamed. Naming is now consistent across the board.	Mixing CamelCase and snake_case in the same script gets confusing fast, especially when sharing code.	Resolved

Summary

8	**5**	**2**	**1**	**309**	**0**
Total Changes Logged	Issues Resolved	Items Verified Clean	Step Skipped	Records Preserved	Rows Deleted

Personal Note
Honestly the dataset was cleaner than I expected — numeric columns were already in the right dtype, and the dates were consistently formatted even though stored as plain strings. Most of the actual work was around text standardisation and the column rename. The CouponCode nulls took a moment to think through — initially considered dropping them but realised they just represent customers who didn't use a coupon, so filling with 'No Coupon' made more sense. Didn't want to lose 309 valid orders over that. Each step was verified before moving on rather than just running code and assuming it worked.